

Syllabus

Spring Semester, 2026

Course Title	Course Code	Type	Credits	Hours
Calculus III	MC3110	Core Requirement	3	4

Instructor	Section	Office	Office Hours	Phone	Email	Classroom
Yongdeok Kim	1,2	H3401	M3 M4 W5	051-606-2282	ydkimjia67@gmail.com	H3402
Dongsoo Lee	1(EC)	H3505	M5 Tu5 W5	061-606-2228	lee.dongsoo@ksa.kaist.ac.kr	H3405
Jongmin Kim	2(EC)	H3507	M6 Tu6 Th6	051-606-2149	kimjongmin@ksa.hs.kr	H3402

1. Course Description

This course provides a rigorous introduction to multivariable calculus and vector analysis, with a particular emphasis on both the theoretical foundations and practical applications. Beginning with the geometry of real-valued functions, limits, and continuity, students will develop the tools necessary to study differentiation in higher dimensions, including gradients, directional derivatives, and optimization techniques such as the method of Lagrange multipliers.

The course then advances to the study of vector fields and integral calculus in two and three dimensions, covering line integrals, surface integrals, and the classical integral theorems of Green, Stokes, and Gauss. Throughout, connections will be made to physics and engineering, particularly in the modeling of motion, forces, and fluxes.

A balance between conceptual understanding and problem-solving is emphasized. Students will be expected to not only carry out computations, but also to interpret and communicate the underlying mathematical structures. By the end of the course, students will have developed a solid foundation in multivariable calculus and its applications, preparing them for further study in mathematics, physics, engineering, and related fields.

2. Learning Outcomes

Core Competencies				
Problem Solving	Reasoning	Communication	Attitude and Practice	Problem Identification and Formulation

3. Textbooks and References

- Primary Textbook: James Stewart et al. – *Calculus*, 7th edition
- Reference Book: Jerold E. Marsden & Anthony Tromba – *Vector Calculus*, 6th edition (Although officially listed as a supplementary reference, this text provides the *conceptual framework of the lectures*.)

4. Teaching Methods

- Lectures on fundamental concepts and main examples
- Student presentations and discussions on selected problems
- Homework assignments and exams for evaluation

5. Evaluation Criteria

a. Components

Assignments & Participation	Midterm	Final Exam	Attendance & Attitude	Total
50	210	280	60	600

b. Methods

1. **Exams (Midterm, Final):** 170 minutes each, closed book, essay/problem-solving format. Partial credit given. No calculators or electronic devices allowed. Students without a basis score will be required to take a Make-up Examination.
2. **Assignments:** About 4 sets. Deadlines announced separately. Late submission penalty: 20% deduction per day (24h). After 5 days (120h), not graded (0 points). Server timestamp determines submission time.
3. **Presentations/Participation:** Evaluated based on presentations, discussions, and engagement.
4. **Attendance and Grading:** Tardiness: -1 point; Absence: -3 points. Grading is based on absolute standards, with possible small adjustments depending on exam difficulty.

6. Grading Policy

- Grades: A(+ / 0 / -), B(+ / 0 / -), C(+ / 0 / -), D(+ / 0 / -), F.
- Grades will be determined primarily by students' demonstrated achievement, with overall class performance also taken into consideration.
- A grade of B or higher indicates strong and satisfactory mastery of the course material.
- Attendance for at least three-quarters of the total class hours is required to earn credit.
- In case of an examination missed due to official leave or medical leave, the score will be calculated by multiplying [the score corresponding to the total percentile score earned by the student on other examinations in the same subject] by [100% for official leave, or 80% for medical leave]. However, if no reference score is available, a make-up examination will be conducted

7. Weekly Schedule

Week (Period)	Topics and Content	Remarks
Week 1 Feb. 23–27	2.1 The Geometry of Real-Valued Functions 2.2 Limits and Continuity	
Core Competency	Reasoning, Problem Identification and Formulation Formulate limit/continuity problems and identify key conditions.	
Week 2 Mar. 2–6	2.3 Differentiation 2.4 Introduction to Paths and Curves 2.5 Properties of the Derivative	
Core Competency	Problem Solving, Reasoning Compute derivatives and justify basic properties of differentiability.	
Week 3 Mar. 9–13	2.6 Gradients and Directional Derivatives 3.1 Iterated Partial Derivatives	
Core Competency	Problem Solving, Communication Compute and interpret gradients and directional derivatives in examples.	
Week 4 Mar. 16–20	3.2 Taylor's Theorem 3.3 Extrema of Real-Valued Functions	
Core Competency	Reasoning, Problem Identification and Formulation Set up Taylor approximation and extrema problems with correct hypotheses.	
Week 5 Mar. 23–27	3.4 Constrained Extrema and Lagrange Multipliers 3.5 The Implicit Function Theorem 4.1 Acceleration and Newton's Second Law	
Core Competency	Problem Solving, Reasoning, Attitude and Practice Solve constrained extrema and implicit-function problems carefully.	
Week 6 Mar. 30 – Apr. 3	4.2 Arc Length 4.3 Vector Fields 4.4 Divergence and Curl	
Core Competency	Problem Identification and Formulation, Communication Model geometric/physical situations using vector fields and related concepts.	
Week 7 Apr. 6 – 10	5.1 Introduction 5.2 The Double Integral Over a Rectangle 5.3 The Double Integral Over More General Regions	
Core Competency	Problem Solving, Reasoning Set up and evaluate double integrals with correct regions and bounds.	
Week 8 Apr. 13–17	Midterm Exam (Apr. 14–17)	Apr. 13 Regular class

Week (Period)	Topics and Content	Remarks
Week 9 Apr. 20–24	5.4 Changing the Order of Integration 5.5 The Triple Integral 6.1 The Geometry of Maps from $\mathbb{R}^2$ to $\mathbb{R}^n$	
Core Competency	Problem Solving, Reasoning Change integration order, evaluate triple integrals, and interpret maps geometrically.	
Week 10 Apr. 27– May 1	6.2 The Change of Variables Theorem 6.3 Applications 6.4 Improper Integrals	
Core Competency	Problem Identification and Formulation, Communication Choose substitutions appropriately and explain Jacobian-based setups clearly.	
Week 11 May 4–8	7.1 The Path Integral 7.2 Line Integrals 7.3 Parametrized Surfaces	
Core Competency	Problem Solving, Reasoning Parametrize curves/surfaces and set up path and line integrals correctly.	
Week 12 May 11–15	7.4 Area of a Surface 7.5 Integrals of Scalar Functions Over Surfaces 7.6 Surface Integrals of Vector Fields	
Core Competency	Problem Solving, Communication Compute surface area/integrals and present parametrization choices clearly.	
Week 13 May 18–22	7.7 Applications to Differential Geometry, Physics, and Forms of Life 8.1 Green's Theorem 8.2 Stokes' Theorem	
Core Competency	Reasoning, Problem Identification and Formulation Identify when Green's/Stokes' theorem applies and set orientation correctly.	
Week 14 May 25–29	8.3 Conservative Fields 8.4 Gauss' Theorem	
Core Competency	Problem Solving, Communication Identify conservative fields and apply Gauss' theorem in context.	
Week 15 June 1–5	8.5 Some Differential Equations of Mechanics and Technology	
Core Competency	Problem Solving, Reasoning Interpret and analyze basic differential-equation models from mechanics and technology.	
Week 16 June 8–12	Makeup Class Period	
Week 17 June 15–19	Final Exam (June 15–19) Chapters 5, 6, 7, 8	

※ The course schedule is subject to change according to the academic calendar.